

Working mechanism for Vamos

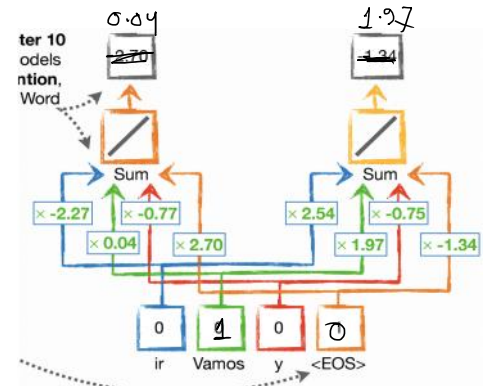
We predicted Vamos as first word.

① Embedding values:

$$E_{\text{Vamos}} = [0.04 \quad 1.97]$$

You can find it by taking one-hot vector over the vocabulary =

$$\begin{bmatrix} \text{in} \\ \text{Vamos} \\ y \\ \text{<EOS>} \end{bmatrix}$$



② Positional Encoding

$$E_{\text{Vamos}} = [0.04 \quad 1.97]$$

Pos = 1 [2nd word in the decoder sentence, Pos=1]

$$i = 0, 0$$

$$\begin{aligned} PE(\text{pos}=1, 2 \times i=0) &= \sin\left(\frac{\text{pos}}{10000^{\frac{2i}{\text{dim}}}}\right) \\ \text{[for } -0.84] &= \sin\left(\frac{1}{10000^{\frac{2 \times 0}{2}}}\right) = 0.84 \quad \text{[in Radian]} \end{aligned}$$

$$\begin{aligned} PE(\text{pos}=2, 2 \times i+1=1) &= \cos\left(\frac{\text{pos}}{10000^{\frac{2i}{\text{dim}}}}\right) \\ \text{[for } 0.54] &= \cos\left(\frac{2}{10000^{\frac{2 \times 0}{2}}}\right) = 0.54 \quad \text{[in Radian]} \end{aligned}$$

Embedding of Vamos after adding P.E.

$$E_{\text{Vamos}} = [0.04 \quad 1.97] + [0.84 \quad 0.54] = [0.88 \quad 2.51]$$

[The slide uses toy values]

$$\text{so } E_{\text{Vamos}} = [0.86 \quad 2.37]$$

③ Calculate self attention of Vamos:

[There is only <EOS> and Vamos here]

The Query, key and value weights

$$Q = \begin{bmatrix} 0.4 & 0.4 \\ -0.3 & 0.1 \end{bmatrix} \quad K = \begin{bmatrix} 0.4 & -0.7 \\ -0.4 & 0.3 \end{bmatrix} \quad V = \begin{bmatrix} -1.1 & -0.7 \\ -0.4 & 1.3 \end{bmatrix}$$

Q, K, V of Vamos

$$E_{\text{Vamos}} = [-0.86 \quad 2.37]$$

$$Q_{\text{Vamos}} = E_{\text{Vamos}} \times Q = [-0.86 \quad 2.37] \times \begin{bmatrix} 0.4 & 0.4 \\ -0.3 & 0.1 \end{bmatrix} \\ = [-1.1 \quad -0.1]$$

$$K_{\text{Vamos}} = E_{\text{Vamos}} \times K = [-0.86 \quad 2.37] \times \begin{bmatrix} 0.4 & -0.7 \\ -0.4 & 0.3 \end{bmatrix} \\ = [-1.3 \quad -0.1]$$

$$V_{\text{Vamos}} = E_{\text{Vamos}} \times V = [-0.86 \quad 2.37] \times \begin{bmatrix} -1.1 & -0.7 \\ -0.4 & 1.3 \end{bmatrix} \\ = [0 \quad 3.7]$$

K, V of <EOS>

$$E_{\text{EOS}} = [2.7 \quad -0.34]$$

[Embedding after positional Encoding, Not the Embedding after attention from previous <EOS> unit at Decoder]

$$K_{\text{EOS}} = E_{\text{EOS}} \times K = [2.7 \quad -0.34] \times \begin{bmatrix} 0.4 & -0.7 \\ -0.4 & 0.3 \end{bmatrix} \\ = [1.2 \quad -1.8]$$

$$V_{\text{EOS}} = E_{\text{EOS}} \times V = [2.7 \quad -0.34] \times \begin{bmatrix} -1.1 & -0.7 \\ -0.4 & 1.3 \end{bmatrix} \\ = [-2.8 \quad -2.3]$$

and to calculate self attention of Vamos, we need Query, key value of Vamos and key value of <EOS>

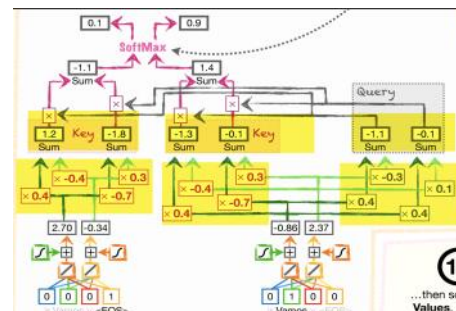
→ Same Query, key, value matrix that was used for <EOS>

Attention of Vamos

① We will multiply Query of Vamos with the key of <EOS> and key of Vamos

$$Q_{Vamos} \cdot K_{EOS} = \begin{bmatrix} -1.1 & -0.1 \end{bmatrix} \cdot \begin{bmatrix} 1.2 & -1.8 \end{bmatrix} \\ = (-1.1) \times 1.2 + (-0.1) \times (-1.8) = -1.1$$

$$Q_{Vamos} \cdot K_{Vamos} = \begin{bmatrix} -1.1 & -0.1 \end{bmatrix} \cdot \begin{bmatrix} -1.3 & -0.1 \end{bmatrix} \\ = (-1.1) \times (-1.3) + (-0.1) \times (-0.1) = 1.4$$



② Do Softmax to find similarity to Vamos

$$\text{Softmax}(Q_{Vamos} \cdot K_{EOS}, Q_{Vamos} \cdot K_{Vamos})$$

$$= \text{softmax}(-1.1, 1.4)$$

$$= \left[\frac{e^{-1.1}}{e^{-1.1} + e^{1.4}}, \frac{e^{1.4}}{e^{-1.1} + e^{1.4}} \right]$$

$$= [0.1, 0.9]$$

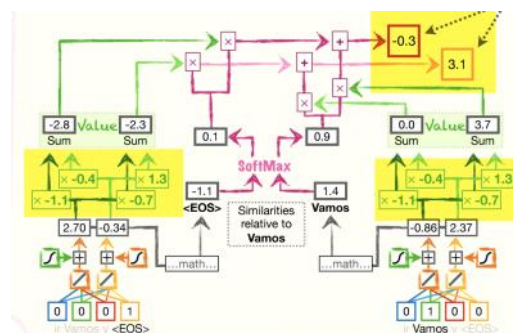
③ Attention for Vamos

$$V_{EOS} \times 0.1, V_{Vamos} \times 0.9$$

$$= \begin{bmatrix} -2.8 & -2.3 \end{bmatrix} \times 0.1 + \begin{bmatrix} 0 & 3.7 \end{bmatrix} \times 0.9$$

$$= \begin{bmatrix} (-2.8 \times 0.1) + 0 \times 0.9 & (-2.3 \times 0.1) + (3.7 \times 0.9) \end{bmatrix}$$

$$= \begin{bmatrix} 0.3 & 3.1 \end{bmatrix}$$

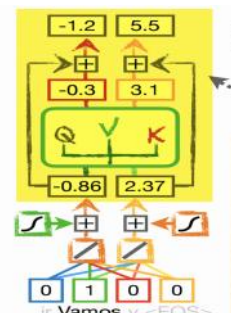


④ Residual connection [update Embedding of Vamos]

$$E_{Vamos} = E_{Vamos} + \text{Attention of Vamos}$$

$$= \begin{bmatrix} -0.86 & 2.37 \end{bmatrix} + \begin{bmatrix} 0.3 & 3.1 \end{bmatrix}$$

$$= \begin{bmatrix} -1.2 & 5.5 \end{bmatrix}$$



④ Calculate Cross Attention for Vamos

To calculate cross attention we need Query & Value of Vamos [Query & Value of decoder token] and Key & Value of Lets and Key & Value of go [Key & Value of encoder tokens]

Q, V of Vamos :

Query & Value weights, $Q = \begin{bmatrix} 1.5 & -0.3 \\ 0.3 & -1.0 \end{bmatrix}$ $V = \begin{bmatrix} 1.1 & 0.6 \\ -1.2 & 0.5 \end{bmatrix}$

$E_{Vamos} = \begin{bmatrix} -1.2 & 5.5 \end{bmatrix}$ [after self attention]

$Q_{Vamos} = E_{Vamos} \times Q = \begin{bmatrix} -1.2 & 5.5 \end{bmatrix} \times \begin{bmatrix} 1.5 & -0.3 \\ 0.3 & -1.0 \end{bmatrix}$
 $= \begin{bmatrix} -0.1 & -5.1 \end{bmatrix}$

$V_{Vamos} = E_{Vamos} \times V = \begin{bmatrix} -1.2 & 5.5 \end{bmatrix} \times \begin{bmatrix} 1.1 & 0.6 \\ -1.2 & 0.5 \end{bmatrix}$
 $= \begin{bmatrix} 2.6 & 1.2 \end{bmatrix}$

K, V of Lets and go: (same as <EOS> side's calculation)

NO NEED TO RECALCULATE!!

key and value weight matrices

$K = \begin{bmatrix} -1.1 & 0.3 \\ -0.3 & -0.8 \end{bmatrix}$ $V = \begin{bmatrix} 1.1 & 0.6 \\ -1.2 & 0.5 \end{bmatrix}$

$E_{Lets} = \begin{bmatrix} 4.37 & -1.01 \end{bmatrix}$ [after self attention at encoder]
 $E_{go} = \begin{bmatrix} 0.82 & -1.43 \end{bmatrix}$

$K_{Lets} = E_{Lets} \times K = \begin{bmatrix} 4.37 & -1.01 \end{bmatrix} \times \begin{bmatrix} -1.1 & 0.3 \\ -0.3 & -0.8 \end{bmatrix} = \begin{bmatrix} -4.5 & 2.1 \end{bmatrix}$

$K_{go} = E_{go} \times K = \begin{bmatrix} 0.82 & -1.43 \end{bmatrix} \times \begin{bmatrix} -1.1 & 0.3 \\ -0.3 & -0.8 \end{bmatrix} = \begin{bmatrix} -0.5 & 1.4 \end{bmatrix}$

$V_{Lets} = E_{Lets} \times V = \begin{bmatrix} 4.37 & -1.01 \end{bmatrix} \times \begin{bmatrix} 1.1 & 0.6 \\ -1.2 & 0.5 \end{bmatrix} = \begin{bmatrix} 6.0 & 3.1 \end{bmatrix}$

$$V_{go} = E_{go} \times V = \begin{bmatrix} 0.82 & -1.43 \end{bmatrix} \times \begin{bmatrix} 1.1 & 0.6 \\ -1.2 & 0.5 \end{bmatrix} = \begin{bmatrix} 2.6 & 1.2 \end{bmatrix}$$

Cross attention

① We will multiply Query of Vamos [decoder] with the key of Lets and key of go [Encoder]

$$Q_{Vamos} \cdot K_{Lets} = \begin{bmatrix} -0.1 & -5.1 \end{bmatrix} \cdot \begin{bmatrix} -4.5 & 2.1 \end{bmatrix}$$

$$= (-0.1) \times (-4.5) + (-5.1) \times 2.1 = -10.3$$

$$Q_{Vamos} \cdot K_{go} = \begin{bmatrix} -0.1 & -5.1 \end{bmatrix} \cdot \begin{bmatrix} -0.5 & 1.4 \end{bmatrix}$$

$$= (-0.1) \times (-0.5) + (-5.1) \times 1.4 = -7.1$$

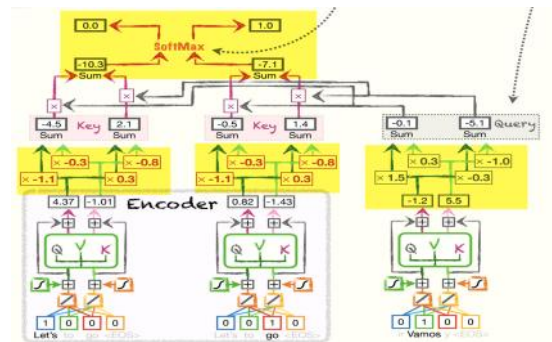
② Do softmax to find similarity to Vamos

$$\text{softmax}(Q_{Vamos} \cdot K_{Lets}, Q_{Vamos} \cdot K_{go})$$

$$= \text{softmax}(-10.3, -7.1)$$

$$= \left[\frac{e^{-10.3}}{e^{-10.3} + e^{-7.1}}, \frac{e^{-7.1}}{e^{-10.3} + e^{-7.1}} \right]$$

$$= [0, 1]$$

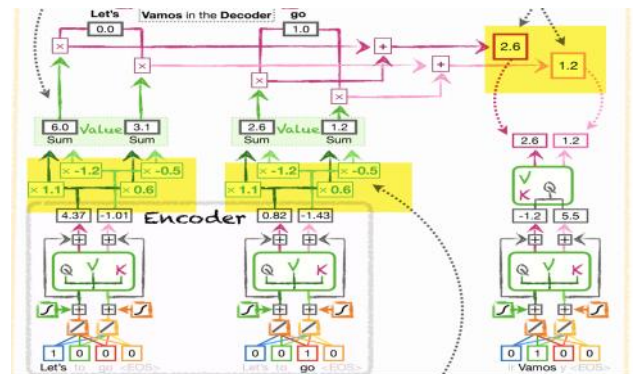


③ Attention for Vamos

$$(V_{Lets} \times 0) + (V_{go} \times 1)$$

$$= \begin{bmatrix} 6.0 & 3.1 \end{bmatrix} \times 0 + \begin{bmatrix} 2.6 & 1.2 \end{bmatrix} \times 1$$

$$= \begin{bmatrix} 2.6 & 1.2 \end{bmatrix}$$

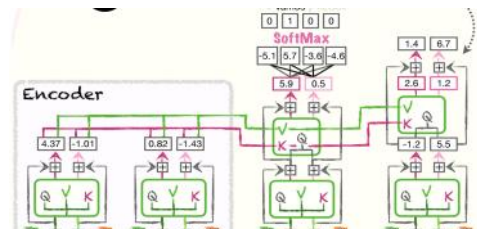


④ Residual connection [update Embedding of Vamos]

$$E_{Vamos} + \text{Attention of Vamos}$$

$$= \begin{bmatrix} 1.2 & 5.5 \end{bmatrix} + \begin{bmatrix} 2.6 & 1.2 \end{bmatrix}$$

$$= \begin{bmatrix} 1.4 & 6.7 \end{bmatrix}$$



$$= \begin{bmatrix} 1.4 & 6.7 \end{bmatrix}$$

⑤ Next token prediction

$\begin{bmatrix} 1.4 & 6.7 \end{bmatrix}$ gets multiplied with a weight matrix and we get $\begin{bmatrix} -14.8 & -3.5 & -10 & 9.8 \end{bmatrix}$

$$\text{Softmax} \left(\begin{bmatrix} -14.8 & -3.5 & -10 & 9.8 \end{bmatrix} \right)$$

$$= \begin{bmatrix} 0 & 0 & 0 & 1 \end{bmatrix}$$

↓

$$\begin{bmatrix} \text{ir} & \text{Vamos} & \text{y} & \text{<EOS>} \end{bmatrix}$$

<EOS> gets predicted as next token.
 so we have reached the end of sentence at the decoder side. No more prediction is needed. Translated sentence 'Vamos'

